

Design Document: ZIP Code 3.0 Application

Project Title

ZIP Code 3.0: Blocked Sequence Set Processor for US Postal Data

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Objective

To design and implement a C++ application that:

1. Converts standard ZIP code CSV data into a length-indicated and blocked sequence set format.
2. Supports efficient insertion, deletion, and searching of records using block buffering and indexing.
3. Provides tools for summary reporting, index building, and structured data visualization.

System Overview

The application processes ZIP code data stored in structured files using metadata headers, block-level records, and simple indexes. It enables:

- Conversion from CSV to length-indicated files.
- Building blocked sequence set (.bss) files.
- Indexing and querying using relative block numbers (RBN).
- Interactive and CLI-based utilities for summary, insertions, deletions, and dumps.

Key Components

1. Data Structures

- **ZipRecord:** Represents a ZIP code with associated fields.
- **Field:** Metadata for a single field in a record (name, type, format, isPrimaryKey).

- **Header:** Metadata for the entire file including structure type, block size, record count, etc.
- **Block:** Represents a block in the sequence set including RBN links and records.
- **AvailBlock:** Metadata for a deleted/unused block.

2. Core Modules

- **CSVBuffer:** Reads, converts, and indexes CSV or length-indicated files.
- **HeaderBuffer / HeaderBuffer3:** Reads and writes structured headers.
- **BlockBuffer:** Packs/unpacks block data to/from disk.
- **RecordBuffer:** Unpacks a record string into fields.
- **DumpUtility:** Dumps blocks in physical or logical order for inspection.

3. Programs

- **main.cpp:** Interactive entry point for CSV conversion and summary generation.
- **InsertRecord.cpp:** Inserts new records into the blocked file.
- **DeleteRecord.cpp:** Deletes records by ZIP key and manages avail list.
- **ConvertToBlocked.cpp:** Converts a length-indicated file to blocked format.
- **BuildIndex.cpp:** Builds a simple key-to-RBN index for fast lookup.
- **SearchIndex.cpp:** Searches the blocked data using the generated index and zip flags.

File Formats

- **CSV:** Flat file with comma-separated fields.
- **Length-Indicated File:** File with a structured header and prefixed record lengths.
- **Blocked Sequence Set (BSS):** Fixed-size blocks containing multiple sorted records.
- **Index File:** Contains ordered pairs of {key, RBN}.

User Interfaces

- **Interactive CLI:** Convert CSV and generate summary.
- **Command-Line Flags:** Support for -z##### ZIP lookups, and file I/O.

Error Handling

- File I/O errors (missing, corrupted, unreadable files).
- Format errors (malformed records, incorrect header structure).
- Logic errors (duplicate keys, overfilled blocks).

Future Enhancements

- Support for variable block sizes.
- Improved compression or binary storage.
- GUI-based interface for visualization.